

APPENDIX C: SERVICE LEVEL AGREEMENTS (SLAS), NETWORK KPIs & LIQUIDATED DAMAGES

5G FWA PROGRAM (5,000 SITES NATIONWIDE) Tender Reference: RFP-TELTECH-FWA-2026-001 Project: Nationwide 5G FWA Project (Band n50, 60 MHz TDD)
Operator: Telecommunications Technology Solutions Incorporated (TelTech)
Author: OLIVER TUNGOL --- ## 1. OBJECTIVE & GOVERNANCE PRINCIPLES

This Appendix defines the mandatory Service Level Agreements (SLAs), Key Performance Indicators (KPIs), measurement methodologies, reporting intervals, and financial remedies (Service Credits and Liquidated Damages) governing:

- 1. Turnkey Deployment & Commissioning Phases (Lots 1, 2, 5)**
- 2. 5G Radio & Core Network Operational Performance (Lots 1, 3)**
- 3. OSS/BSS Platform Reliability & Field Dispatch Operations (Lot 4)**
- 4. Managed Services & First Line Maintenance Operations (Lot 6)** Performance shall be evaluated on a calendar monthly basis using the automated performance management OSS and audited ticketing records. ---
2. PROJECT ROLLOUT & DELIVERY SLAS ### 2.1 Milestone Delivery Schedule

The Contractor shall adhere strictly to the following turnkey delivery schedule for Phase 1 (1,000 sites):

- Milestone 1 (TSSR & Nominal Engineering Approval):** Within **45 Calendar Days** of contract award for all 1,000 priority sites.
- Milestone 2 (First 200 Sites Commercial On-Air):** Within **120 Calendar Days** of contract award.
- Milestone 3 (Cumulative 500 Sites Commercial On-Air):** Within **210 Calendar Days** of contract award.
- Milestone 4 (Cumulative 1,000 Sites Commercial On-Air & Integrated):** Within **365 Calendar Days** of contract award. **### 2.2 Delay Liquidated Damages (Rollout)**
- If the Contractor fails to achieve Final Acceptance of a milestone by the scheduled completion date, the Operator shall deduct **Delay Liquidated Damages (DLD)** equal to **0.5% of the total milestone contract value per calendar week of delay**, up to a maximum aggregate cap of **10.0% of the total contract value**.
- Delays exceeding 60 calendar days beyond the milestone date shall entitle the Operator to terminate for default and draw down on the Contractor's Performance Bond. --- **## 3. RADIO ACCESS NETWORK (RAN) OPERATIONAL KPIs & SERVICE CREDITS ### 3.1 Core Radio KPIs (Calculated 24/7/365 Nationwide)**

Radio Availability (%) = [(Total Operating Cell Hours - Total Unplanned Outage Cell Hours) / Total Operating Cell Hours] * 100

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| KPI-RAN-01 | Radio Network Availability | Monthly aggregate across all active cells | >= 99.95% | 99.7
| KPI-RAN-02 | 5G Session Setup Success (CSSR) | Initial RRC connection + 5G QoS flow setup | >= 99.60% |
| KPI-RAN-03 | 5G Session Drop Rate (CDDR) | Abnormal session drops during active data sessions | <= 0.1
| KPI-RAN-04 | Peak Sector DL Throughput | Busy hour average across cell cluster | >= 450 Mbps | 380 Mbp
| KPI-RAN-05 | Cell-Edge DL User Speed | 95th percentile stationary CPE speed | >= 50 Mbps | 25 Mbps | M
| KPI-RAN-06 | Cell-Edge UL User Speed | 95th percentile stationary CPE upload | >= 15 Mbps | 5 Mbps | U
| KPI-RAN-07 | Radio Round-Trip Latency (RTT) | CPE to local gNodeB/UPF edge | <= 18 ms | 30 ms | Audit p
| KPI-RAN-08 | Packet Loss Rate (PLR) | User plane packet discard rate | <= 0.01% | 0.10% | Transport &
| :--- | :--- | :---: | :---: | :--- |
| KPI-5GC-01 | Control Plane NF Availability (AMF/SMF/UDM) | >= 99.999% ("Five Nines") | Monthly | \10,0
| KPI-5GC-02 | User Plane (UPF) Packet Availability | >= 99.999% | Monthly | \10,000USD$ per hour outage
| KPI-5GC-03 | PDU Session Establishment Success Rate | >= 99.80% | 24-hr Rolling | Root-cause analysis
| KPI-5GC-04 | CG-NAT Port Allocation Success Rate | >= 99.99% | Monthly | \5,000USD$ penalty for subscri
| KPI-5GC-05 | Active-Active Datacenter Failover Time | <= 30Seconds | Semi-annual Drill | Penalty of 5%
| KPI-5GC-06 | UPF Internal Packet Latency | <= 1.0 ms | Continuous | Hardware NIC DPDK tuning | --- ##
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| KPI-BSS-01 | Converged Charging System (CHF) Uptime | >= 99.99% | 2.0% monthly BSS maintenance fee ded
| KPI-BSS-02 | Top-Up Processing Speed (GCash / Maya) | <= 3.0Seconds | Mandatory optimization of API ga
| KPI-OSS-01 | New Installation Dispatch Lead Time | <= 48Hours | >= 95% of orders completed within 48h
| KPI-OSS-02 | First-Time-Right (FTR) Installation Rate | >= 96.0% | Repeat truck rolls at Contractor exp
| KPI-OSS-03 | Churned CPE Hardware Recovery Rate | >= 70.0% | Contractor debited \55USD$ per unrecovere
| KPI-OSS-04 | Automated Work-Order Dispatch Engine Capacity | >= 3,500orders/day | Zero backlog accumula
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| Severity Level | Definition / Operational Impact | Response SLA | MTTR SLA (Urban) | MTTR SLA (Rural)
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| Severity 1 | Complete macro site outage; Core NF failure; | <= 15 Mins | <= 2.0 Hours | <= 4.0 Hours |
| (Critical) | >500 subscribers out of service; major backhaul | | |
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| Severity 2 | Single sector down; degraded throughput; | <= 30 Mins | <= 4.0 Hours | <= 6.0 Hours |
| (Major) | loss of DC battery backup; generator fault | | |
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| Severity 3 | Non-service affecting hardware fault; loss of | <= 1 Hour | <= 12.0 Hours | <= 24.0 Hours
| (Minor) | redundancy (single PSU failed, SFP degraded) | | |
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| Severity 4 | Configuration inquiry, routine audit, planned | <= 4 Hours | <= 48.0 Hours | <= 72.0 Hour
| (Warning/Info) | preventative maintenance scheduling | | |
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For every hour (or fraction thereof) that a Severity 1 incident exceeds the mandated MTTR threshold, the Operator shall deduct:

- **Urban / Suburban Sites: 0.5% of monthly site Managed Services fee per hour** of prolonged outage.
- **Rural Sites: 0.25% of monthly site Managed Services fee per hour** of prolonged outage.
- Total monthly service credit deductions for Managed Services shall be capped at **20.0% of the total monthly Managed Services fee**. --- ## 7. MONTHLY REPORTING & GOVERNANCE CADENCE 1. **Daily Operational Flash Report:** Generated automatically at 06:00 PHT summarizing the previous 24 hours of network availability, outages, MTTR compliance, and work orders.
- 2. **Weekly Engineering Review:** Joint operations meeting reviewing open trouble tickets, cluster drive tests, and rollout milestone progress.
- 3. **Monthly Governance & SLA True-Up:** Formal executive session reviewing all KPI metrics against SLA thresholds, finalizing service credits, and approving contractor invoices. ---

